

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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System Name	Koperasi Kakitangan KADA SDN.BHD
Group Name	DATA ALCHEMIST

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

*<< Provide the SQL statements (DDL and DML) for a set of queries (minimum **10** queries) that can be implemented in your proposed module. In addition, please include the screenshots of the output/results for each SQL statement.>>*

3.2 Interface Implementation

*<< Provide at least **THREE (3)** interface design reflecting queries in Subsection 3.1.>>*

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

<< Provide achievement(s) that have been achieved in your module and suggestion(s) to further improvise in future .>>

4.2 Reflection

<< Provide your reflection based on your WBL project .>>

3. SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 A) 2 SQL (DDL) statements

1. Create table member

```
mysql> CREATE TABLE Member(  
  -> no_anggota VARCHAR (255) PRIMARY KEY,  
  -> nama VARCHAR (255) NOT NULL,  
  -> no_kp VARCHAR (255) NOT NULL,  
  -> no_pf VARCHAR (255) UNIQUE NOT NULL,  
  -> no_telefon VARCHAR (255),  
  -> emel VARCHAR (255)  
  -> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> describe Member;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| no_anggota | varchar(255)  | NO   | PRI | NULL    |       |  
| nama       | varchar(255)  | NO   |     | NULL    |       |  
| no_kp      | varchar(255)  | NO   |     | NULL    |       |  
| no_pf      | varchar(255)  | NO   | UNI | NULL    |       |  
| no_telefon | varchar(255)  | YES  |     | NULL    |       |  
| emel       | varchar(255)  | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
6 rows in set (0.00 sec)
```

2. Create table guarantorReplacement

```
mysql> CREATE TABLE GuarantorReplacement (  
  -> no_GRF VARCHAR(255) PRIMARY KEY,  
  -> no_loan VARCHAR(255) NOT NULL,  
  -> no_kp_Pbaru VARCHAR(255) NOT NULL,  
  -> nama_Pbaru VARCHAR(255) NOT NULL,  
  -> no_anggota_Pbaru VARCHAR(255) NOT NULL,  
  -> no_pf_Pbaru VARCHAR(255) NOT NULL,  
  -> no_kp_saksi VARCHAR(255) NOT NULL,  
  -> nama_saksi VARCHAR(255) NOT NULL,  
  -> no_anggota_saksi VARCHAR(255) NOT NULL,  
  -> no_pf_saksi VARCHAR(255) NOT NULL,  
  -> statusReplacement VARCHAR(255),  
  -> suratPengakuanPemohon VARCHAR(255),  
  -> noID_clerk VARCHAR(255),  
  -> no_anggota VARCHAR(255),  
  -> FOREIGN KEY (no_anggota) REFERENCES Member(no_anggota)  
  -> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> describe GuarantorReplacement;
```

Field	Type	Null	Key	Default	Extra
no_GRF	varchar(255)	NO	PRI	NULL	
no_loan	varchar(255)	NO		NULL	
no_kp_Pbaru	varchar(255)	NO		NULL	
nama_Pbaru	varchar(255)	NO		NULL	
no_anggota_Pbaru	varchar(255)	NO		NULL	
no_pf_Pbaru	varchar(255)	NO		NULL	
no_kp_saksi	varchar(255)	NO		NULL	
nama_saksi	varchar(255)	NO		NULL	
no_anggota_saksi	varchar(255)	NO		NULL	
no_pf_saksi	varchar(255)	NO		NULL	
statusReplacement	varchar(255)	YES		NULL	
suratPengakuanPemohon	varchar(255)	YES		NULL	
noID_clerk	varchar(255)	YES		NULL	
no_anggota	varchar(255)	YES	MUL	NULL	

```
14 rows in set (0.00 sec)
```

B) 8 SQL (DML) statements, with at least 3 JOIN queries related to the proposed module

1. Insert data into Member table

```
mysql> SELECT * FROM Member;
```

no_anggota	nama	no_kp	no_pf	no_telefon	emel
M001	Ali Ahmad	900101-01-1234	PF001	0123456789	ali@gmail.com
M002	Ahmad Saiful	921201-02-2345	PF002	0198765432	ahmad@gmail.com
M003	Noraini Jamal	800501-01-5678	PF003	0171122334	noraini@gmail.com

```
3 rows in set (0.00 sec)
```

2. Insert data into GuarantorReplacement table

```
mysql> INSERT INTO GuarantorReplacement (
  -> no_GRF, no_loan, no_kp_Pbaru, nama_Pbaru, no_anggota_Pbaru, no_pf_Pbaru, no_kp_saksi, nama_saksi, no_anggota_saksi, no_pf_saksi, statusReplacement, suratPengakuanPemohon, no_anggota)
  -> VALUES
  -> ('GRF001', 'LN001', '921201-02-2345', 'Ahmad Saiful', 'M002', 'PF002', '800501-01-5678', 'Noraini Jamal', 'M003', 'PF003', 'Pending', 'AcknowledgmentLetter1.pdf', 'M001'),
  -> ('GRF002', 'LN002', '890101-02-7890', 'Farid Aziz', 'M004', 'PF004', '920301-01-8901', 'Sarah Ismail', 'M005', 'PF005', 'Approved', 'AcknowledgmentLetter2.pdf', 'M002');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO GuarantorReplacement (
  -> no_GRF, no_loan, no_kp_Pbaru, nama_Pbaru, no_anggota_Pbaru, no_pf_Pbaru, no_kp_saksi, nama_saksi, no_anggota_saksi, no_pf_saksi, statusReplacement, suratPengakuanPemohon, no_anggota)
  -> VALUES
  -> ('GRF003', 'LN003', '001023-10-5643', 'Yusuf Arman', 'M005', 'PF005', '920420-14-8764', 'Alya Sofea', 'M006', 'PF006', 'Pending', 'AcknowledgmentLetter3.pdf', 'M003');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM GuarantorReplacement;
```

no_GRF	no_loan	no_kp_Pbaru	nama_Pbaru	no_anggota_Pbaru	no_pf_Pbaru	no_kp_saksi	nama_saksi	no_anggota_saksi	no_pf_saksi	statusReplacement	suratPengakuanPemohon	noID_clerk	no_anggota
GRF001	LN001	921201-02-2345	Ahmad Saiful	M002	PF002	800501-01-5678	Noraini Jamal	M003	PF003	Approv	AcknowledgmentLetter1.pdf	NULL	M001
GRF002	LN002	890101-02-7890	Farid Aziz	M004	PF004	920301-01-8901	Sarah Ismail	M005	PF005	Approv	AcknowledgmentLetter2.pdf	NULL	M002
GRF003	LN003	001023-10-5643	Yusuf Arman	M005	PF005	920420-14-8764	Alya Sofea	M006	PF006	Pendin	AcknowledgmentLetter3.pdf	NULL	M003

```
3 rows in set (0.00 sec)
```

3. Update Status of a Guarantor Replacement

```
mysql> UPDATE GuarantorReplacement
-> SET statusReplacement = 'Pending'
-> WHERE no_GRF = 'GRF001';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM GuarantorReplacement;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| no_GRF | no_loan | no_kp_Pbaru | nama_Pbaru | no_anggota_Pbaru | no_pf_Pbaru | no_kp_saksi | nama_saksi | no_anggota_saksi | no_pf_saksi | status |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| GRF001 | LN001   | 921201-02-2345 | Ahmad Saiful | M002             | PF002       | 800501-01-5678 | Noraini Jamal | M003             | PF003       | Pendin |
| GRF002 | LN002   | 890101-02-7890 | Farid Aziz   | M004             | PF004       | 920301-01-8901 | Sarah Ismail  | M005             | PF005       | Approv |
| GRF003 | LN003   | 001023-10-5643 | Yusuf Arman  | M005             | PF005       | 920420-14-8764 | Alya Sofea    | M006             | PF006       | Pendin |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4. Retrieve Members with Pending Guarantor Replacements

```
mysql> SELECT Member.nama AS MemberName, GuarantorReplacement.statusReplacement
-> FROM Member
-> JOIN GuarantorReplacement ON Member.no_anggota = GuarantorReplacement.no_anggota
-> WHERE GuarantorReplacement.statusReplacement = 'Pending';
+-----+-----+
| MemberName | statusReplacement |
+-----+-----+
| Ali Ahmad   | Pending           |
| Noraini Jamal | Pending           |
+-----+-----+
2 rows in set (0.00 sec)
```

5. List Guarantors and Witnesses for All Loans

```
mysql> SELECT GuarantorReplacement.no_loan, GuarantorReplacement.nama_Pbaru AS GuarantorName, GuarantorReplacement.nama_saksi AS WitnessName
-> FROM GuarantorReplacement;
+-----+-----+-----+
| no_loan | GuarantorName | WitnessName |
+-----+-----+-----+
| LN001   | Ahmad Saiful  | Noraini Jamal |
| LN002   | Farid Aziz    | Sarah Ismail  |
| LN003   | Yusuf Arman   | Alya Sofea    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

6. Count Pending Guarantor Replacements for Each Member

```
mysql> SELECT Member.nama AS MemberName, COUNT(*) AS PendingRequests
-> FROM Member
-> JOIN GuarantorReplacement ON Member.no_anggota = GuarantorReplacement.no_anggota
-> WHERE GuarantorReplacement.statusReplacement = 'Pending'
-> GROUP BY Member.nama;
+-----+-----+
| MemberName | PendingRequests |
+-----+-----+
| Ali Ahmad   | 1               |
| Noraini Jamal | 1               |
+-----+-----+
2 rows in set (0.00 sec)
```

7. Retrieve All Approved Guarantor Replacements

```
mysql> SELECT *
-> FROM GuarantorReplacement
-> WHERE statusReplacement = 'Approved';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| no_GRF | no_loan | no_kp_Pbaru | nama_Pbaru | no_anggota_Pbaru | no_pf_Pbaru | no_kp_saksi | nama_saksi | no_anggota_saksi | no_pf_saksi | statusRe |
| lacement | suratPengakuanPemohon | noID_clerk | no_anggota | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| GRF002 | LN002 | 890101-02-7890 | Farid Aziz | M004 | | PF004 | | 920301-01-8901 | Sarah Ismail | M005 | | PF005 | | Approved |
| | | AcknowledgmentLetter2.pdf | NULL | M002 | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8. Retrieve members with guarantor replacements.

```
mysql> SELECT Member.nama AS MemberName, GuarantorReplacement.no_loan, GuarantorReplacement.statusReplacement
-> FROM Member
-> INNER JOIN GuarantorReplacement
-> ON Member.no_anggota = GuarantorReplacement.no_anggota;
+-----+-----+-----+
| MemberName | no_loan | statusReplacement |
+-----+-----+-----+
| Ali Ahmad | LN001 | Pending |
| Ahmad Saiful | LN002 | Approved |
| Noraini Jamal | LN003 | Pending |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

3.2 Interface Implementation

1. GuarantorReplacement table



**SENARAI PEMOHONAN
PENUKARAN PENJAMIN**

Tapis Status

Tapis Saksi

Tapis Penjamin

Tapis No. Loan









No. GRF	GRF 001	GRF 002	GRF 003
No. loan	LN001	LN002	LN003
No. KP Penjamin Baru	921201-02-2345	890101-02-7890	001023-10-5643
Nama Penjamin Baru	Ahmad Saiful	Farid Aziz	Yusuf Arman
No. Anggota Penjamin Baru	M002	M004	M005
No. PF Penjamin Baru	PF002	PF004	PF005
No. KP Saksi	800501-01-5678	920301-01-8901	920420-14-8764
Nama Saksi	Noraini Jamal	Sarah Ismail	Alya Sofea
No. Anggota Saksi	M003	M005	M006
No. PF Saksi	PF003	PF005	PF006
Surat Pengakuan Pemohon	AcknowledgementLetter1.pdf	AcknowledgementLetter2.pdf	AcknowledgementLetter3.pdf
Status Penukaran	Pending	Approved	Pending
No. ID Kerani	NULL	NULL	NULL
No. Anggota	M001	M002	M003

2. Members with Pending Guarantor Replacements



**SENARAI PEMOHONAN
PENUKARAN PENJAMIN**

Tapis Status

Tapis Saksi

Tapis Penjamin

Tapis No. Loan

 Pending







Member Name	Status Replacement
Ali Ahmad	Pending
Noraini Jamal	Pending

3. Count Pending Guarantor Replacements for Each Member



**SENARAI PEMOHONAN
PENUKARAN PENJAMIN**

Tapis Status

 Pending

Tapis Saksi



Tapis Penjamin



Tapis No. Loan



Member Name	Pending Request
Ali Ahmad	1
Noraini Jamal	1

4. List of Guarantors and Witnesses for All Loans



**SENARAI PEMOHONAN
PENUKARAN PENJAMIN**

Tapis Status



Tapis Saksi

 Ya

Tapis Penjamin

 Ya

Tapis No. Loan

 Ya

No Loan	Guarantor Name	Witness Name
LN001	Ahmad Saiful	Noraini Jamal
LN002	Farid Aziz	Sarah Ismail
LN003	Yusuf Arman	Alya Sofea

4. SECTION D (CONCLUSION AND REFLECTION)

4.1 CONCLUSION

The Guarantor Replacement Module fixes a serious problem with the loan application procedure, greatly improving the KADA Cooperative System's functionality. This module makes sure that in case of unexpected occurrences like the loss of a guarantor, members can easily change their guarantor details. By including this feature, the system shows a strong dedication to flexibility and member support, which eventually raises customer satisfaction.

Then, the module also makes sure that the procedure is safe and easy to use by involving both members and clerks. To preserve the integrity of the system, clerks check and acknowledge the guarantor replacement requests that members submit through online. Next, the notification via email to member could increase the communication between the company and members.

Therefore, by offering a simple and effective method for updating guarantor details, this module improves user experience overall and guarantees that members feel supported during critical times and this type of flexibility not only meets members' changing needs but also shows how dedicated the system is to help its users.

4.2 REFLECTION

1. Description (Pure fact)

The WBL project is a long-term initiative, taking nearly two semesters to complete. In Year 1, Semester 2, we began this project in our System Analysis and Design class. The process started with a meeting with our client, KADA Cooperative System, followed by group discussions to analyze the client's requirements. Once we identified the client's needs, we proposed a system designed to digitalize their manual processes, aligning with the ongoing technological advancements worldwide. This digital transformation ensures that we stay competitive and up to date.

To propose the system effectively, we focused on thorough analysis by preparing detailed reports, designing ERDs, and creating interface prototypes using Figma. Given the broad scope of the project, tasks were divided equally among the team members. For instance, there were nine processes to complete, with each team member handling 2-3 processes from start to finish. This approach ensured that everyone gained the same level of knowledge and contributed equally to the project.

In Year 2, Semester 1, we continued the project by bringing it to life. This phase of the project was integrated into three subjects: System Development and Technology, Database, and Software Engineering. During this time, we implemented what we had learned into the system and prepared to present it to our client at the end of the semester.

2. Descriptions (Feelings)

At the beginning of the project, I was quite excited about working on it, especially since only Data Engineering students get the opportunity to work with a real client. This added a sense of responsibility and seriousness to the project. However, I faced several challenges throughout the process. For example, I had additional responsibilities with JKM KTDI, which sometimes affected my studies. To overcome this, I had to manage my time effectively to ensure I completed the assigned tasks within the given timeframe.

Fortunately, having unlimited internet access allowed me to find information about things I didn't know. This made it easier for me to gather resources and learn independently, as everything is readily available at our fingertips.

3. Evaluation

One of the positive aspects of the project experience was working with my groupmates. Initially, we were not very close and only interacted for the sake of the project. However, as time went by, we became much closer and started helping each other to complete tasks together.

On the other hand, a challenging part of the experience was the tendency to compare our work with other groups. This often made us feel like we were falling behind. However, this negative experience motivated us to improve our system and strive to catch up, ensuring that we were not left behind.

These experiences and emotions significantly influenced my behavior, making me more productive and instilling a sense of urgency in completing tasks efficiently.

If there are some issues in our group, we would say it directly to that person, so that he/she is aware of their behavior.

4. Analysis

I chose to highlight this incident because people often don't realize their mistakes unless we point them out. We cannot assume they will acknowledge their wrongdoings on their own, as humans tend to believe they are always right. The preceding three stages make sense, as we first discovered the purpose of our project, then reflected on our feelings, and finally evaluated it in relation to our project.

The main area of concern throughout this project is that teamwork is crucial. Without effective communication with our groupmates, completing the project would have been significantly more challenging.

5. Conclusion

After completing this project, I have gained insights into how a cooperative company operates as a subsidiary of a larger organization. I also learned how the loan process works and why people choose to take loans despite the interest or benefit percentages they need to pay. Additionally, I developed valuable skills in working collaboratively with others. Working on this project not only opened my eyes to the realities of the professional world but also enhanced both my technical and soft skills.

One lingering question I have is whether KADA will fully utilize the system we worked on for nearly a year. I sincerely hope they make good use of it, as everyone in this course has truly given their best effort to this project.

6. Self-criticism

I will manage my time better, as I feel that my packed schedule may sometimes cause the workflow to slow down. Therefore, moving forward, I plan to organize my time more effectively by listing out my tasks and prioritizing the most urgent and important ones first. I also realize that I need moral support from others to help me stay motivated and focused, as I have been struggling to complete my work.